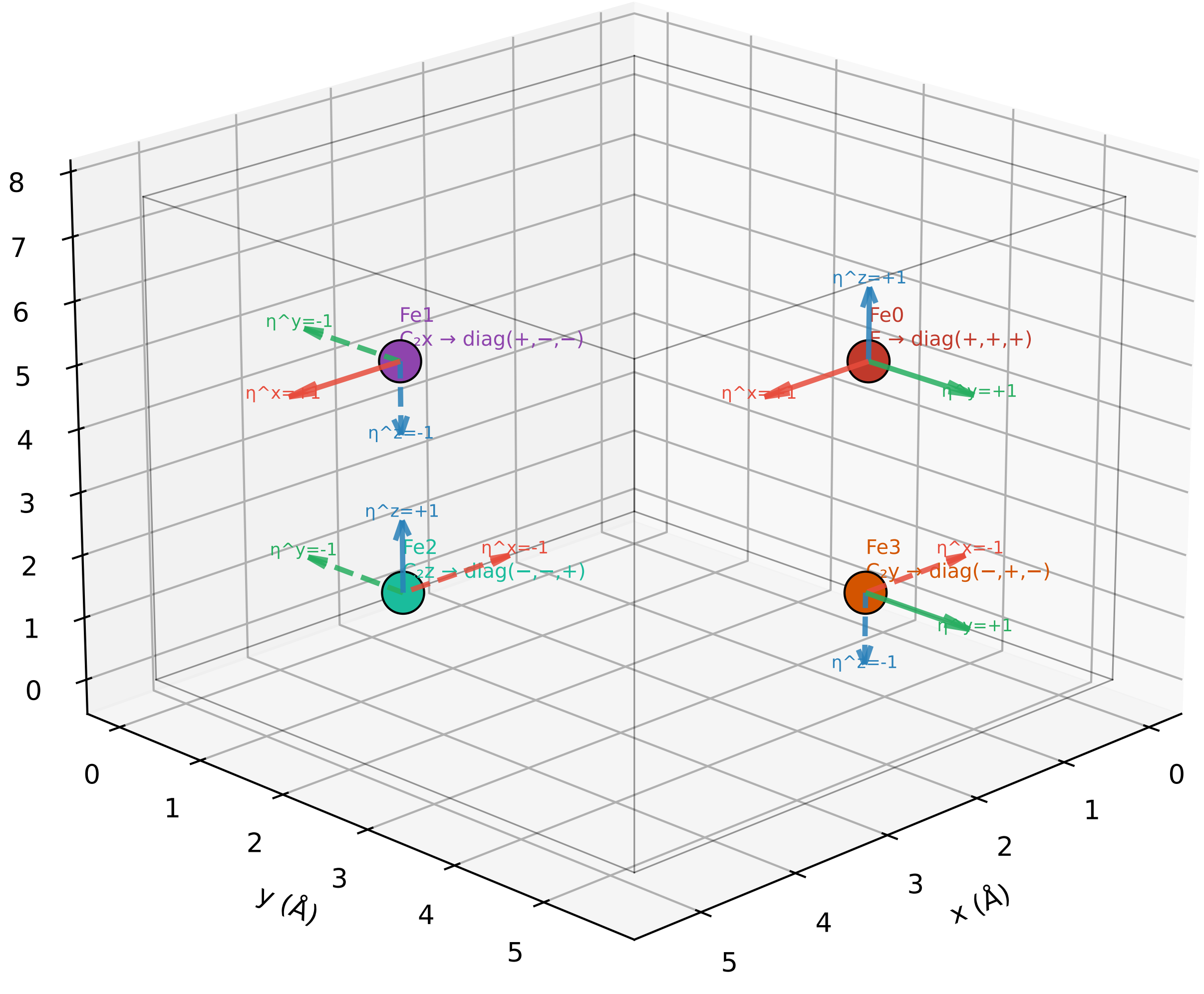
**Fe sublattice local frames (R_fe_frames in unitcell_builders.cpp)**
**Arrows: η^x (red), η^y (green), η^z (blue). Solid = +1, dashed = −1**

Fe frame diagonal elements = rotated Fe spin components
H[λ²]_j += R^(k)_αα · χ₂α · S^α_Fe,k → η^α_Fe = R^(k)_αα

Fe local frame arrows at each sublattice position




| Component | sub 0 | sub 1 | sub 2 | sub 3 | Pattern |
|---|---|---|---|---|---|
| η^x (R_xx) | +1 | +1 | −1 | −1 | σ_A |
| η^y (R_yy) | +1 | −1 | −1 | +1 | σ_C |
| η^z (R_zz) | +1 | −1 | +1 | −1 | σ_G |

```
CODE convention sigma vectors:
  σ_F = (+,+,+,+)  ferromagnetic
  σ_G = (+,−,+,−)  G-type AFM  ← Γ₂ ordering
  σ_A = (+,+,−,−)  A-type AFM
  σ_C = (+,−,−,+)  C-type AFM

Dot products (key for selection rules):
  σ_G · σ_G = 4   σ_G · σ_A = 0
  σ_G · σ_C = 0   σ_G · σ_F = 0
  σ_A · σ_F = 0   σ_C · σ_F = 0
```